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$$\sum_{n=2}^{\infty} \frac{\ln n}{n}$$

→ integraal criterium

$$\lim_{t \rightarrow +\infty} \int_2^t \frac{\ln x}{x} dx$$

$$t = \ln x \Rightarrow dt = \frac{1}{x} dx$$

$$\Rightarrow \lim_{p \rightarrow +\infty} \int_2^p t dt = \lim_{p \rightarrow +\infty} \left[\frac{t^2}{2} \right]_2^p$$

$$= \lim_{p \rightarrow +\infty} \left[\frac{\ln^2 x}{2} \right]_2^p = +\infty \Rightarrow \text{divergent}$$

References